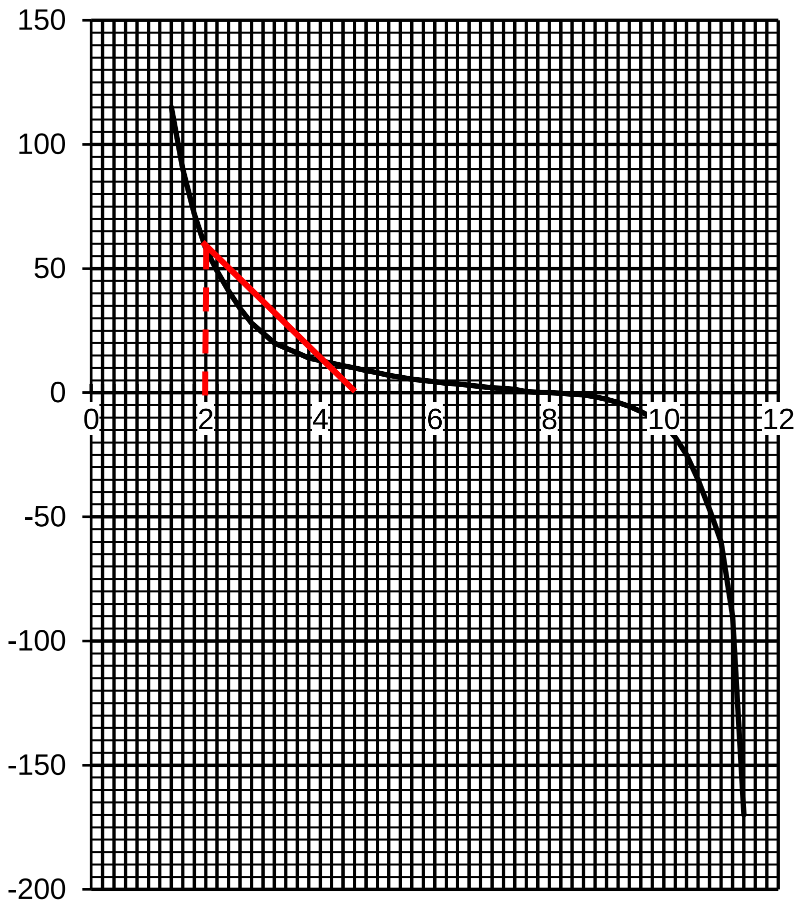
		answer
3(a)	The electric field strength in both sphere A (between $x = 0$ and $x = 1.4$ cm) and sphere B is zero (between $x = 11.4$ and $x = 12.0$ cm).	[1]
3(b)(i)	The resultant field strength is zero at a point between the spheres and this shows that electric fields are in opposite directions in the region between the two spheres. This shows that the polarity of the two charges are the same. Hence, since sphere A is positively charged, sphere B must also be positively charged.	[1] expl [1] state

3(b)(ii)	At x = 0.08 m, the electric field strength due to sphere A cancels out the electric field strength due to sphere B. $E_A = E_B$ $\dfrac{Q_A}{4\pi\varepsilon_o(0.08)^2} = \dfrac{Q_B}{4\pi\varepsilon_o(0.04)^2}$ $\dfrac{Q_A}{Q_B} = \left(\dfrac{0.08}{0.04}\right)^2 = 4$	[1] sub [1] ans
3(c)(i)	The electric field strength is <u>negative of the electric potential gradient</u>, i.e. $E = -\dfrac{dV}{dx}$	[1] don't accept $E = -\dfrac{dV}{dx}$
3(c)(ii)	 From $E = -\dfrac{dV}{dx} \Rightarrow \Delta V = -\int_{x=2cm}^{x=8cm} E\ dx$ Hence, change in potential, ΔV = negative of area under E−x graph (from x = 2.0 cm to x = 8.0 cm) By approximation, area of triangle ≈ area under E−x graph (from x = 2.0 cm to x = 8.0 cm)	[1] ΔV = area under E−x graph [1] value of ΔV

	$\Delta V = -\frac{1}{2}\left(2.6\times10^{-2}\right)\left(60\times10^{6}\right) = -7.8\times10^{5}\ \text{V}$ Work done $=\left(-7.8\times10^{5}\right)\left(-2.0\times10^{-6}\right) = 1.6\ \text{J}$	[1] ans accept 1.4 J to 1.8 J
4(a)	The magnetic flux linkage is the product of the number of turns	[1]